

# ALZ-Foresight: Predicting Alzheimer’s Disease Progression Modeling Using Multimodal Machine Learning on the ADNI Dataset

Ibrahim Labib, Islam Akram, Youssef Elhelw, Mohamed Khattab, and Abdelrahman Elsherbiny  
for the Alzheimer’s Disease Neuroimaging Initiative  
School of Computing and Digital Technology  
ESLSCA University  
Giza, Egypt

**Abstract**—Alzheimer’s disease (AD) is a progressive neurodegenerative disorder that requires reliable early detection and clinically useful prediction systems. This paper presents *ALZ-Foresight*, a multimodal machine learning framework for ADNI-based Alzheimer modeling. The implemented system integrates structured clinical ADNI data with MRI-based image modeling through a CNN branch and a late-fusion prediction strategy. The structured branch uses six ADNI tables, applies subject-level splitting before preprocessing to prevent data leakage, and produces 111 processed clinical, diagnostic, demographic, and genetic features. The earlier structured benchmark showed that XGBoost achieved 92.18% test accuracy and 0.9836 ROC-AUC, while the tuned Top-15 XGBoost model achieved 91.90% test accuracy, 0.9189 weighted F1, and 0.9783 weighted one-vs-rest ROC-AUC. The CNN image branch achieved 70.47% holdout accuracy and 0.8506 ROC-AUC, while the multimodal late-fusion model achieved 73.15% accuracy and 0.8254 ROC-AUC. The deployed web system further passed 19/19 functional and non-functional test cases, including authentication, prediction inference, visualization, explainability, logging, model management, and security checks. These results demonstrate a reproducible multimodal Alzheimer stage-classification system that combines leakage-safe tabular learning, CNN-based MRI modeling, and operational web deployment.

**Index Terms**—Alzheimer’s disease, ADNI, multimodal machine learning, CNN, XGBoost, late fusion, feature selection, leakage prevention

## I. INTRODUCTION

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder characterized by continuing decline in memory, cognition, daily functioning, and independence. Because the disease unfolds gradually and heterogeneously, clinically useful machine learning systems must address both present disease state and future progression risk. Recent work on AD has therefore shifted from static diagnosis toward longitudinal and prognosis-oriented modeling [1].

The Alzheimer’s Disease Neuroimaging Initiative (ADNI) is one of the most influential repositories of data pertaining to Alzheimer’s. This is because it provides standardized longitudinal data across cognitive assessments, diagnosis tables, demographics, genetics, and imaging. However, many research papers cite recurring weaknesses such as leakage through non-subject-level splitting, redundant and highly correlated

variables, uncontrolled missingness, cohort-specific generalization, and limited interpretability [10]. At the same time, the field has diversified. Temporal graph learning has been used for progression forecasting [1]. Transformer-based and deep survival frameworks have been proposed for longitudinal prognosis [8]. Multimodal conversion prediction has been studied in mild cognitive impairment (MCI) [9]. Explainability has also become a major requirement for clinical trust [13].

Recent imaging-based studies continue to show the strength of deep learning. CNN-based systems and transfer-learning frameworks have produced strong MRI classification results [2]. Multimodal deep learning has also improved robustness by combining imaging with clinical or genetic information [7]. Yet highly accurate image-only systems often remain difficult to interpret, computationally expensive, and hard to generalize across settings [13].

Motivated by these trends, this paper introduces *ALZ-Foresight*, a research and implementation framework designed around subject-level split integrity, reproducibility, feature selection, multimodal modeling, and web-based prediction deployment. The implemented system evaluates a structured multi-table ADNI branch, a CNN-based MRI image branch, and a late-fusion multimodal branch for three-class Alzheimer stage classification (DX\_CLEAN: CN, MCI, Dementia/AD). The novelty lies in a leakage-safe, reproducible multi-table ADNI pipeline extended with CNN-based imaging and fusion-based prediction. Specifically, the work includes:

- a leakage-safe visit-level ADNI pipeline that integrates multiple structured tables and enforces subject-level splitting prior to preprocessing;
- a CNN-based MRI image branch for Alzheimer stage classification from representative imaging data;
- a multimodal late-fusion model that combines structured and image-based prediction outputs; and
- a deployed web application with authentication, prediction, visualization, explainability, logging, model versioning, and security validation.

The remainder of this paper is organized as follows. Section II reviews relevant work and synthesizes the literature

used in this study. Section III presents the ALZ-Foresight framework. Section IV describes the data pipeline, multimodal components, preprocessing, and experimental setup. Section V reports the model results. Section VI presents the web-system validation results. Section VII discusses the findings, and Section VIII concludes the paper.

## II. RELATED WORK

All the papers and documents centered around ADNI that were used in the study can be organized under four broad categories: structured and interpretable baselines, MRI-based deep learning, multimodal and temporal modeling, and explainability. Table I summarizes representative studies and their reported results.

### A. Structured and Interpretable Baselines

Structured machine learning remains highly relevant when the inputs are tabular, clinically interpretable, and collected at scale. De Palma et al. [3] showed that very high prognostic performance can be achieved through an XGBoost-Cox survival pipeline using a compact multimodal feature set. Lee et al. [9] found Gradient Boosting Machine (GBM) to be the most robust model for MCI-to-AD conversion when combining demographics, MRI, and PET. Zarei et al. [18] further demonstrated that statistical multimodal predictive markers remain useful when interpretability is a priority. These works support the use of strong structured baselines before moving to heavier end-to-end multimodal architectures.

### B. MRI-Based Deep Learning Baselines

Deep learning has substantially advanced AD classification in imaging-centered settings. Chamundeeswari et al. [2] used CNN-based MRI feature extraction for stage classification. El-Assy et al. [5] achieved high multi-class MRI accuracy using a dual-CNN architecture, while Pandey et al. [11] demonstrated strong transfer-learning performance across ADNI and OASIS. Rehman et al. [12] focused on hippocampus-centered transfer learning, and Ali et al. [20] showed that combining ResNet101 and Mixup augmentation improves class-imbalanced MRI classification. Shukla et al. [6] also showed that preprocessing and dimensionality reduction strongly influence performance in image-based classification pipelines. Although these approaches are powerful, many are unimodal and therefore depend heavily on controlled imaging conditions [5].

### C. Multimodal, Temporal, and Heterogeneity-Aware Models

Recent work increasingly reflects the longitudinal and heterogeneous nature of AD. Aqil et al. [1] used temporal graph learning to model progression, while Machado Reyes et al. [4] introduced subtype-oriented multimodal transformer modeling to explicitly address disease heterogeneity. Karaman et al. [8] applied deep survival analysis to longitudinal multimodal data, and Zhang et al. [19] discovered disease states and progression pathways using personalized hidden-state modeling. Wang et al. [16] showed that multimodal prediction can support cognitive-decline-aware trial design, and a separate study by

Wang et al. [17] enhanced long-term progression prediction by incorporating interaction-aware multimodal fusion. Muksimova et al. [14] demonstrated the importance of multimodal fusion and longitudinal learning in diagnosis settings, and Gupta et al. [7] emphasized that rigorous implementation can cause multimodal systems to outperform isolated branches.

### D. Explainability and Standardization

As performance improves and systems become more complex, interpretability and reproducibility become vital. Saif et al. [13] reviewed explainable AI (XAI) techniques for Alzheimer detection and argued that transparency is becoming nearly as important as raw accuracy. Odusami et al. [10] concluded through systematic review and meta-analysis that multimodal neuroimaging is highly promising, but that heterogeneity still affects the field in preprocessing, validation, and evaluation. These findings matter directly for ALZ-Foresight because the present work prioritizes leakage prevention, subject-level splitting, reproducibility, structured reporting, CNN-based imaging, fusion, and system validation.

### E. Research Gap

The literature reveals four gaps that motivate this study. First, strong results often remain bounded to ADNI-like conditions and do not necessarily generalize [10]. Second, many highly accurate studies still focus on static classification rather than clinically meaningful progression-aware modeling [1]. Third, multimodal gains can be undermined by poor synchronization, weak fusion, or leakage [9]. Fourth, explainability and deployment realism remain underdeveloped relative to raw predictive performance [13]. ALZ-Foresight addresses these issues by combining a leakage-safe structured branch, an MRI-based CNN branch, a late-fusion multimodal model, and an operational web interface with validation across functional and non-functional requirements.

TABLE I: Representative studies used in the literature review

| Ref  | Model       | Focus              | Result       | Limitation           |
|------|-------------|--------------------|--------------|----------------------|
| [1]  | TGN         | Progression        | AUC 0.80     | Low interpretability |
| [2]  | CNN         | MRI classification | High acc     | MRI-only             |
| [3]  | XGB-Cox     | Prognosis          | C-index 0.92 | ADNI bias            |
| [4]  | Transformer | Subtypes           | AUROC 0.73   | No ext. validation   |
| [5]  | Dual-CNN    | Early AD           | 99.6% acc    | MRI-only             |
| [6]  | CNN/RF/XGB  | Classification     | 97.6% acc    | Binary setup         |
| [8]  | Transformer | Prognosis          | AUROC 92.5%  | Label noise          |
| [9]  | GBM/RF/SVM  | Conversion         | AUC 0.875    | Small dataset        |
| [14] | FusionNet   | Multimodal         | 94% acc      | High complexity      |
| [17] | DISFC       | MCI to AD          | AUC 0.96     | Small cohort         |

## III. ALZ-FORESIGHT FRAMEWORK

The framework separates subject-safe data preparation from modality-specific feature extraction, fusion, prediction, and downstream clinical support. This separation is intentional: the structured branch operates on multi-table ADNI clinical data, while the image branch uses MRI inputs processed through a CNN-based classifier. The final multimodal branch combines prediction-level information from both branches using late fusion.

As illustrated in Fig. ??, the ALZ-Foresight architecture organizes the pipeline into sequential components, starting from multimodal data inputs and progressing through feature extraction, classification, fusion, calibration, explainability, and dashboard-based decision support. The figure shows the full intended architecture and now corresponds to the implemented system components more closely because the current system includes both a structured clinical branch and a CNN-based MRI branch. Forecasting and deeper longitudinal survival modeling remain planned extensions beyond the current classification experiments.

The validated structured branch uses visit-level ADNI records and predicts a three-class target (DX\_CLEAN: CN, MCI, Dementia/AD). To prevent subject leakage, all structured splitting is performed by RID before downstream preprocessing. The CNN branch evaluates MRI-based classification into CN, MCI, and AD classes, while the fusion branch combines model outputs using a late-fusion strategy. The final deployed web application exposes the prediction system through a Node.js/Express backend, MongoDB Atlas storage, a Python model inference layer, risk visualization, and AI-generated clinical insights.

#### IV. DATA PIPELINE AND EXPERIMENTAL SETUP

This section describes the validated multimodal branch of ALZ-Foresight. It covers the structured ADNI source tables, inclusion and exclusion criteria, leakage-safe splitting strategy, MRI sample usage, CNN modeling, late fusion, preprocessing decisions, and system deployment setup used for model development and evaluation.

##### A. Dataset Scope, Inclusion, and Exclusion Criteria

The structured experiments use six ADNI tables merged at visit level: ADNIMERGE, Dementia\_Rating, Diagnostic\_Summary, Cognitive\_Scores, PTDEMOG, and ApoE\_Genotyping. Inclusion into the modeling cohort required a valid subject identifier (RID), a normalized visit code, and a resolved diagnostic target mapped to DX\_CLEAN. Demographic and APOE information were linked at subject level, while diagnosis- and cognition-related information were linked at visit level. Table II summarizes the structured ADNI sources used in the validated branch.

TABLE II: ADNI source tables used in the structured branch

| Table              | Role                                |
|--------------------|-------------------------------------|
| ADNIMERGE          | Core visit-level anchor table       |
| Dementia_Rating    | Dementia severity ratings           |
| Diagnostic_Summary | Diagnostic outcomes and labels      |
| Cognitive_Scores   | Neurocognitive assessment variables |
| PTDEMOG            | Subject-level demographics          |
| ApoE_Genotyping    | APOE-related genetic variables      |

Exclusion was performed in three stages for the structured branch. First, identifiers and direct leakage-prone fields were removed before modeling. Second, features with training-partition missingness greater than 0.45 were excluded. Third,

the structured branch retained cognitive, diagnostic, demographic, and genetic variables for tabular model development. For the image branch, representative MRI samples were used for CNN-based classification, as illustrated in Fig. 2.

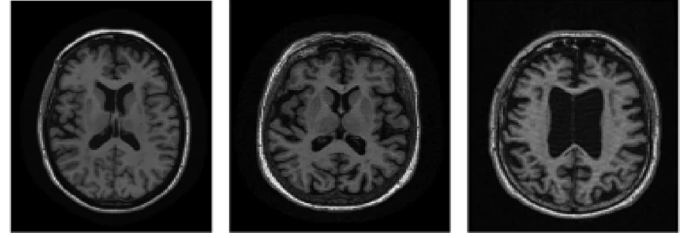


Fig. 2: Representative MRI samples corresponding to cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer's disease (AD), shown from left to right.

##### B. Leakage Prevention, Splitting Strategy, and Class Structure

To ensure leakage-safe evaluation, the structured train/validation/test split was performed by RID before any transformation. This is essential in ADNI-style longitudinal data, where row-wise splitting can leak subject-specific information across partitions. The resulting row-level and RID-level sizes are shown in Table III. At row level, the split is approximately 70/15/15.

TABLE III: Leakage-safe RID-based split sizes

| Partition  | Rows | Unique RIDs |
|------------|------|-------------|
| Train      | 8004 | 1686        |
| Validation | 1714 | 361         |
| Test       | 1740 | 362         |

The target variable is DX\_CLEAN, a three-class label representing cognitively normal (CN), mild cognitive impairment (MCI), and Dementia/AD. The CNN and fusion evaluations report the equivalent class structure as CN, MCI, and AD. Class imbalance remains present in the validation and test partitions, with MCI constituting the largest class. For that reason, weighted F1, macro F1, balanced accuracy, Matthews correlation coefficient (MCC), and ROC-AUC were reported in addition to raw accuracy.

##### C. Preprocessing, Missingness Control, and Feature Space

After excluding identifiers and leakage-prone variables, missingness was computed on the training partition only. A threshold of 0.45 removed 139 columns and produced a final processed feature space of 111 variables. These 111 features include 88 numeric and 23 categorical variables. No correlation-based pruning or PCA-based dimensionality reduction was used in the final reported structured branch; redundancy control was instead deferred to training-only missingness filtering followed by model-based feature importance. This decision preserves feature traceability and supports interpretability. Some features, including CDRSB and MMSE, are closely related to diagnostic criteria; future work will evalu-

ate model performance after excluding such proxy variables. Table IV summarizes the feature processing stage.

TABLE IV: Feature processing summary

| Item                         | Value |
|------------------------------|-------|
| Missingness threshold        | 0.45  |
| Dropped columns              | 139   |
| Remaining processed features | 111   |
| Numeric features             | 88    |
| Categorical features         | 23    |

#### D. CNN Branch and Late Fusion

The MRI branch implements a CNN-based image classifier for the same three diagnostic classes: CN, MCI, and AD. The CNN branch produces class probabilities for each image input. These probability outputs are then combined with the structured branch outputs using a late-fusion approach. In the final fusion experiment, the CNN branch was assigned a weight of 0.75, and the resulting fused prediction was evaluated on a holdout set of 745 samples.

This design implements multimodal learning in two forms. First, the structured branch integrates multiple clinical sources, including cognitive, diagnostic, demographic, and genetic information. Second, the overall system combines this structured information with MRI-derived CNN predictions through late fusion. This makes the current version of ALZ-Foresight a true multimodal implementation rather than only a conceptual multimodal design.

#### E. Experimental Setup and Reproducibility

The structured branch was executed in a notebook-based Python environment using pandas, numpy, scikit-learn, xgboost, catboost, and imbalanced-learn. The earlier benchmark phase compared XGBoost, CatBoost, and logistic regression. The updated branch retrained XGBoost on the 111-feature processed dataset, extracted model-based feature importances, selected the top 15 features, and applied randomized hyperparameter search.

The CNN branch was evaluated on a separate holdout result set using CN, MCI, and AD labels. The fusion branch combined the CNN prediction outputs with the structured model outputs using late fusion. The deployed system uses a Node.js/Express backend, MongoDB Atlas, a Python inference layer, and AI-generated clinical insights through GPT-4o. The model inference endpoint returns the predicted class, confidence score, and class probability distribution, while the frontend renders risk visualization and explanation panels.

### V. MODELING AND EXPERIMENTAL RESULTS

This section reports the structured tabular results, CNN image-branch results, and multimodal late-fusion results. The structured branch establishes the strongest tabular baseline, while the CNN and fusion branches demonstrate the final multimodal extension.

#### A. Earlier Structured Benchmark Phase

The earlier benchmark phase compared three structured-data classifiers on preprocessed train/validation/test CSV files: XGBoost, CatBoost, and logistic regression (SGD-based). XGBoost achieved the best test performance, followed by CatBoost, with a substantial gap over the linear baseline. The results are reported in Table V. These findings indicate that the structured ADNI feature space contains nonlinear interactions better captured by tree-based ensembles than by a linear model.

TABLE V: Earlier structured benchmark phase results

| Model               | Acc.   | Prec.  | Rec.   | F1     | ROC-AUC |
|---------------------|--------|--------|--------|--------|---------|
| XGBoost             | 0.9218 | 0.9224 | 0.9218 | 0.9219 | 0.9836  |
| CatBoost            | 0.9109 | 0.9113 | 0.9109 | 0.9110 | 0.9824  |
| Logistic Regression | 0.7661 | 0.7780 | 0.7661 | 0.7663 | –       |

#### B. 111-Feature Structured Model

The first model in the updated multi-table branch used all 111 processed features. This model reached 88.27% validation accuracy and 90.92% test accuracy, with weighted F1 scores of 0.8830 and 0.9088, respectively. These results confirm that the leakage-safe multi-table pipeline already supports strong stage prediction. However, they also raise the question of whether the full feature set contains unnecessary redundancy.

#### C. Feature Importance Analysis

Feature importance extracted from the 111-feature XGBoost model showed that cognitive and functional severity variables dominate prediction. The most influential features were CDRSB, CDRSB\_bl, FAQ, PTCOGBEG, mPACctrailsB, and LDELTOTAL\_BL. The presence of PTCOGBEG suggests that symptom-duration information contributes meaningful signal, while the appearance of VISDATE\_ptd indicates a temporal-drift component inside the structured branch. APOE-related variables remain present, but at lower relative importance than the top cognitive measures. The top 20 features are listed in Table VI; the first 15 were used in the reduced models.

TABLE VI: Top 20 features ranked by XGBoost importance

| Rank | Feature         | Importance |
|------|-----------------|------------|
| 1    | CDRSB           | 0.1997     |
| 2    | CDRSB_bl        | 0.0818     |
| 3    | FAQ             | 0.0480     |
| 4    | PTCOGBEG        | 0.0367     |
| 5    | mPACctrailsB    | 0.0270     |
| 6    | ORIGPROT        | 0.0233     |
| 7    | LDELTOTAL_BL    | 0.0209     |
| 8    | mPACCdigit_bl   | 0.0198     |
| 9    | PTADDX          | 0.0143     |
| 10   | mPACCdigit      | 0.0140     |
| 11   | MMSE_bl         | 0.0136     |
| 12   | mPACctrailsB_bl | 0.0122     |
| 13   | FSVERSION_bl    | 0.0096     |
| 14   | VISDATE_ptd     | 0.0094     |
| 15   | ADAS13          | 0.0092     |
| 16   | PHASE_ptd       | 0.0085     |
| 17   | USERDATE_ptd    | 0.0080     |
| 18   | PTGENDER_ptd    | 0.0079     |
| 19   | APOE4           | 0.0078     |
| 20   | APOE4_count     | 0.0076     |

#### D. Reduced Top-15 Structured Model

To further improve efficiency and reduce redundancy, the feature space was reduced to the top 15 most important features identified by the XGBoost model. This compact representation resulted in improved generalization performance within the updated structured branch.

The untuned Top-15 XGBoost model achieved a validation accuracy of 88.91% and a test accuracy of 91.55%, with corresponding weighted F1 scores of 0.8893 and 0.9154. Hyperparameter tuning was then applied using randomized search, producing the final tuned Top-15 model. The tuned model achieved 91.90% test accuracy, 0.9189 weighted F1, 0.9181 macro F1, 0.9193 balanced accuracy, 0.8735 MCC, 0.2642 log loss, and 0.9783 weighted one-vs-rest ROC-AUC.

For the tuned Top-15 model, the test class-wise results were: CN ( $P = 0.93, R = 0.95, F1 = 0.94$ ), Dementia ( $P = 0.90, R = 0.91, F1 = 0.91$ ), and MCI ( $P = 0.92, R = 0.90, F1 = 0.91$ ). The confusion matrices in Fig. 3 and Fig. 4 show that most remaining errors occur between MCI and adjacent classes, which is clinically expected.

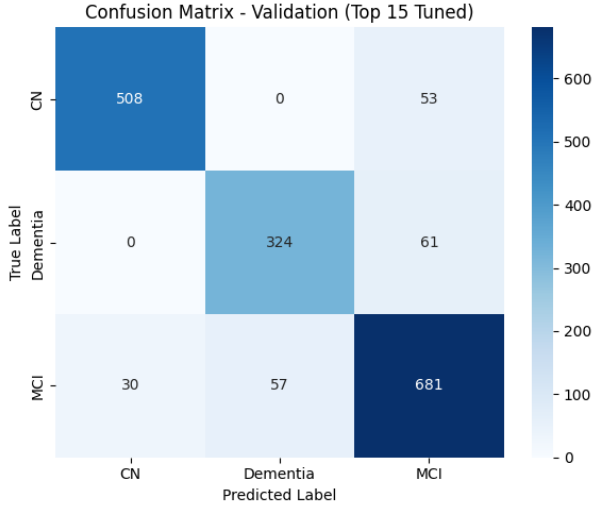


Fig. 3: Validation confusion matrix for the tuned Top-15 XGBoost model. Most residual errors occur between MCI and Dementia, while CN remains comparatively well separated.

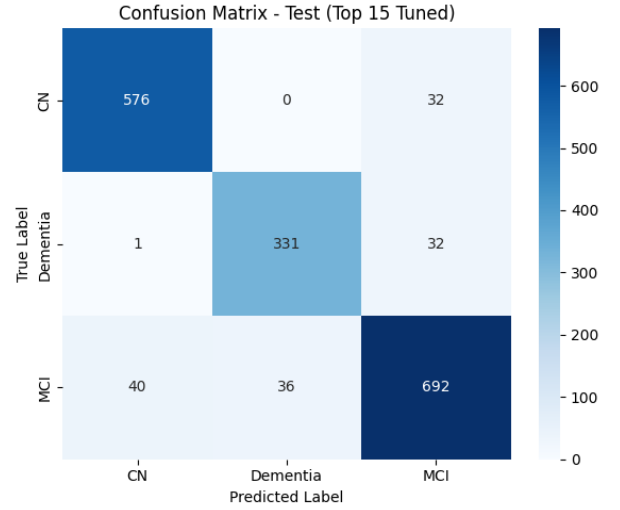


Fig. 4: Test confusion matrix for the tuned Top-15 XGBoost model. The test set shows stronger CN separation and balanced performance across all three classes.

#### E. CNN and Multimodal Fusion Results

The CNN image branch was evaluated on a holdout set of 745 samples. It achieved 70.47% accuracy, 0.7046 weighted F1, 0.6869 macro F1, and 0.8506 ROC-AUC. The late-fusion model combined the CNN and structured outputs using a CNN weight of 0.75. This fused model achieved 73.15% accuracy, 0.7305 weighted F1, 0.6787 macro F1, and 0.8254 ROC-AUC. Table VII compares the structured, CNN, and fusion branches.

TABLE VII: Comparison of structured, CNN, and fusion branches

| Branch               | Accuracy | Weighted F1 | Macro F1 | ROC-AUC |
|----------------------|----------|-------------|----------|---------|
| Tuned Top-15 XGBoost | 0.9190   | 0.9189      | 0.9181   | 0.9783  |
| CNN image branch     | 0.7047   | 0.7046      | 0.6869   | 0.8506  |
| Late-fusion model    | 0.7315   | 0.7305      | 0.6787   | 0.8254  |

Table VIII reports the class-level CNN and fusion results. The CNN branch produced relatively balanced CN and MCI performance, while the AD class remained more difficult due to lower support. The late-fusion model improved CN recall substantially and increased overall accuracy, but AD precision remained the weakest class-level outcome.

TABLE VIII: Class-level CNN and fusion performance on holdout set

| Branch | Class | Precision | Recall | F1     |
|--------|-------|-----------|--------|--------|
| CNN    | CN    | 0.7157    | 0.7157 | 0.7157 |
| CNN    | MCI   | 0.7127    | 0.7167 | 0.7147 |
| CNN    | AD    | 0.6374    | 0.6237 | 0.6304 |
| Fusion | CN    | 0.7853    | 0.8930 | 0.8357 |
| Fusion | MCI   | 0.7747    | 0.6431 | 0.7028 |
| Fusion | AD    | 0.4554    | 0.5484 | 0.4976 |

The holdout confusion matrices for the CNN and fusion branches are summarized in Table IX. The fusion model improved CN classification substantially, increasing CN true positives from 214 to 267. However, the shift also increased MCI-to-AD and MCI-to-CN confusion, showing that fusion improved overall accuracy but still requires additional calibration and class-balancing work.

TABLE IX: CNN and fusion confusion matrices on holdout set

| Branch | Actual | Pred. CN | Pred. MCI | Pred. AD |
|--------|--------|----------|-----------|----------|
| CNN    | CN     | 214      | 76        | 9        |
| CNN    | MCI    | 76       | 253       | 24       |
| CNN    | AD     | 9        | 26        | 58       |
| Fusion | CN     | 267      | 30        | 2        |
| Fusion | MCI    | 67       | 227       | 59       |
| Fusion | AD     | 6        | 36        | 51       |

## VI. SYSTEM IMPLEMENTATION AND VALIDATION

The ALZ-Foresight web application was implemented as a local development system using a Node.js/Express backend, MongoDB Atlas database, Python-based model inference, and AI-generated clinical insight generation. The test scope covered authentication and security, patient management, AI prediction, visualization, reliability, logging, and model management. The system passed all 19 test cases, as summarized in Table X.

TABLE X: System validation summary

| Category                       | Tests   | Result |
|--------------------------------|---------|--------|
| Authentication & Security      | 3 / 3   | PASS   |
| Patient Management             | 3 / 3   | PASS   |
| Input Validation               | 1 / 1   | PASS   |
| AI Prediction                  | 2 / 2   | PASS   |
| Visualization & Explainability | 3 / 3   | PASS   |
| Audit & Logging                | 2 / 2   | PASS   |
| Model Management               | 2 / 2   | PASS   |
| Non-Functional Requirements    | 3 / 3   | PASS   |
| Total                          | 19 / 19 | PASS   |

The prediction endpoint returns a diagnosis label, confidence score, and full class probability distribution. In the tested example, the endpoint returned an MCI prediction with 0.7928 confidence and class probabilities for CN, Dementia, and MCI. Direct XGBoost model inference completed in 649 ms, which is below the 3 s tabular prediction requirement. The full API response, including AI-generated insights, took approximately 6.4 s. The system also handled OpenAI API failure gracefully by returning the core prediction even if the insight-generation step failed.

The frontend includes risk visualization through diagnosis badges, confidence percentages, and probability bars for CN,

MCI, and Dementia. Explainability is implemented through three structured AI-generated explanation panels: clinical interpretation of the result, reasoning behind the prediction, and actionable patient recommendations. Security tests confirmed correct rejection of forged JWT tokens, NoSQL injection attempts, privilege escalation attempts, and unauthenticated access. Reliability tests showed that five concurrent requests were handled successfully without instability.

## VII. DISCUSSION

The results support four main conclusions.

First, the structured XGBoost branch remains the strongest individual predictive component. The earlier benchmark phase already showed that gradient boosting clearly outperformed a linear baseline, and the updated multi-table branch confirms that the structured ADNI feature space contains nonlinear interactions and threshold effects better captured by tree ensembles. This is consistent with prior structured and prognostic work that found boosting-based methods robust for AD modeling [3].

Second, model-based feature selection improved the structured branch. Although the earlier benchmark phase reported the numerically highest structured result, the reduced Top-15 branch is valuable because it substantially compresses the input space while preserving competitive performance. Within the updated 111-feature branch itself, the reduced models improved from 90.92% test accuracy to 91.55% for the untuned Top-15 model and 91.90% for the tuned Top-15 model. This indicates that feature-importance-driven dimensionality reduction reduced redundancy and improved generalization in the new pipeline.

Third, the CNN and fusion results demonstrate that ALZ-Foresight now includes implemented multimodal functionality, not only a planned multimodal architecture. The CNN branch produced meaningful image-based classification performance with 70.47% holdout accuracy and 0.8506 ROC-AUC. The late-fusion model improved accuracy to 73.15%, indicating that multimodal integration can improve decision-level classification under the tested configuration. However, the fusion ROC-AUC was lower than the CNN-only ROC-AUC, and AD precision remained limited, suggesting that future fusion work should explore calibration, class weighting, stronger MRI preprocessing, and alternative fusion strategies.

Fourth, the system-level tests show that ALZ-Foresight is not only a model-development notebook but also an operational prediction application. The application passed all 19 validation tests, including authentication, RBAC, input validation, prediction inference, timeout handling, visualization, explainability, audit logging, model versioning, model switching, security, and reliability. This strengthens the project by connecting model results to a deployable clinical-support workflow.

The results should still be interpreted carefully. The current implementation performs stage classification rather than complete survival-style time-to-event progression modeling. The title and architecture frame ALZ-Foresight as progression-

oriented because the system is designed for future longitudinal forecasting, longitudinal visualization, and progression-aware extensions. However, the currently reported quantitative model results are stage-classification results for CN, MCI, and AD. Future work should therefore extend the current multimodal classification framework into explicit longitudinal progression prediction.

### VIII. CONCLUSION AND FUTURE WORK

This paper presented ALZ-Foresight, a multimodal machine learning framework for Alzheimer modeling on ADNI. The system integrates a leakage-safe structured ADNI branch, a CNN-based MRI branch, a late-fusion multimodal prediction branch, and a deployed web application for clinical-style prediction, visualization, and explanation.

The structured branch achieved the strongest performance, with the tuned Top-15 XGBoost model reaching 91.90% test accuracy, 0.9189 weighted F1, and 0.9783 weighted one-vs-rest ROC-AUC. The CNN branch achieved 70.47% holdout accuracy and 0.8506 ROC-AUC, while the late-fusion model achieved 73.15% accuracy and 0.8254 ROC-AUC. These results show that structured clinical ADNI variables remain highly predictive, while MRI-based CNN outputs can be incorporated into a multimodal system through late fusion.

The system validation results further showed that the implemented web application passed 19/19 test cases across authentication, patient management, input validation, prediction inference, visualization, explainability, logging, model management, security, and reliability. Future work will focus on stronger MRI preprocessing, improved CNN training, calibrated fusion, alternative early- and late-fusion strategies, external validation, proxy-feature ablation, and explicit longitudinal progression modeling beyond static stage classification.

### ACKNOWLEDGMENT

Data used in this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The ADNI was launched in 2003 as a public-private partnership led by Michael W. Weiner, MD, to develop biomarkers for Alzheimer's disease progression.

Data collection and sharing for this project was funded by the Alzheimer's Disease Neuroimaging Initiative (National Institutes of Health Grant U19AG024904).

### REFERENCES

- [1] K. H. Aqil, P. Dumpuri, K. Ram, and M. Sivaprakasam, "Predictive modeling of Alzheimer's disease progression: Integrating temporal clinical factors and outcomes in time series forecasting," *Intelligence-Based Medicine*, vol. 10, p. 100159, 2024. doi: 10.1016/j.ibmed.2024.100159.
- [2] V. V. Chamundeeswari *et al.*, "Brain MRI analysis using CNN-based feature extraction and machine learning techniques to diagnose Alzheimer's disease," in *Proc. 2024 Conf.*, pp. 526–532, 2024. doi: 10.1109/icdcc62744.2024.10961923.
- [3] L. De Palma *et al.*, "A machine learning pipeline for prognostic modeling of Alzheimer's disease using multimodal data," *Sensors*, vol. 26, no. 5, p. 1523, 2026. doi: 10.3390/s26051523.
- [4] D. Machado Reyes *et al.*, "Identifying progression-specific Alzheimer's subtypes using multimodal transformer," *J. Personalized Medicine*, vol. 14, no. 4, p. 421, 2024. doi: 10.3390/jpm14040421.
- [5] A. M. El-Assy *et al.*, "A novel CNN architecture for accurate early detection and classification of Alzheimer's disease using MRI data," *Scientific Reports*, vol. 14, no. 1, 2024. doi: 10.1038/s41598-024-53733-6.
- [6] G. P. Shukla *et al.*, "Diagnosis and detection of Alzheimer's disease using learning algorithms," *Big Data Mining and Analytics*, vol. 6, no. 4, pp. 504–512, 2023. doi: 10.26599/bdma.2022.9020049.
- [7] M. Gupta *et al.*, "Adversarial network-based classification for Alzheimer's disease using multimodal brain images," *IEEE Access*, 2024. doi: 10.1109/access.2024.3381956.
- [8] B. K. Karaman *et al.*, "A deep survival model for predicting Alzheimer's diagnosis based on multi-modal longitudinal data," *Big Data Mining and Analytics*, vol. 9, no. 2, pp. 465–480, 2026. doi: 10.26599/bdma.2025.9020064.
- [9] M.-W. Lee *et al.*, "A multimodal machine learning model for predicting dementia conversion in Alzheimer's disease," *Scientific Reports*, vol. 14, no. 1, 2024. doi: 10.1038/s41598-024-60134-2.
- [10] M. Odusami *et al.*, "Machine learning with multimodal neuroimaging data to classify stages of Alzheimer's disease: a systematic review and meta-analysis," *Cognitive Neurodynamics*, 2023. doi: 10.1007/s11571-023-09993-5.
- [11] P. K. Pandey *et al.*, "Enhancing healthcare recommendation: Transfer learning in deep CNNs for Alzheimer disease detection," *Frontiers in Medicine*, vol. 11, 2024. doi: 10.3389/fmed.2024.1445325.
- [12] S. U. Rehman *et al.*, "AI-based tool for early detection of Alzheimer's disease," *Heliyon*, vol. 10, no. 8, p. e29375, 2024. doi: 10.1016/j.heliyon.2024.e29375.
- [13] F. H. Saif *et al.*, "Explainable AI for Alzheimer detection: A review," *Applied Sciences*, vol. 14, no. 22, p. 10121, 2024. doi: 10.3390/app142210121.
- [14] S. Muksimova *et al.*, "Multi-modal fusion and longitudinal analysis for Alzheimer's disease classification," *Diagnostics*, vol. 15, no. 6, p. 717, 2025. doi: 10.3390/diagnostics15060717.
- [15] S. K. Kim *et al.*, "Multimodal 3D deep learning for early diagnosis of Alzheimer's disease," *IEEE Access*, 2024. doi: 10.1109/access.2024.3381862.
- [16] C. Wang *et al.*, "A multimodal deep learning approach for prediction of cognitive decline," *Translational Psychiatry*, vol. 14, no. 1, pp. 1–11, 2024. doi: 10.1038/s41398-024-02819-w.
- [17] Y. Wang *et al.*, "Predicting long-term progression of Alzheimer's disease using multimodal deep learning," *J. Translational Medicine*, vol. 22, no. 1, 2024. doi: 10.1186/s12967-024-05025-w.
- [18] S. Zarei *et al.*, "Statistical method for identification of Alzheimer's disease in MCI," *Basic and Clinical Neuroscience*, vol. 16, no. 2, 2025. doi: 10.32598/bcn.2024.2034.7.
- [19] S. Zhang *et al.*, "Machine learning on longitudinal multi-modal data enables prognosis of Alzheimer's disease progression," *iScience*, vol. 27, no. 7, p. 110263, 2024. doi: 10.1016/j.isci.2024.110263.
- [20] R. E. Ali *et al.*, "Refined Alzheimer's disease classification using MRI-based deep learning," in *IMSA*, pp. 569–574, 2024. doi: 10.1109/imsa61967.2024.10652832.